Notes: Bahaj & Reis (RES, 2026)

Jumpstarting an International Currency

Contents

1 Big picture **3**

2 Data and measurement (swap lines, RMB payments, and funding costs) **3**

2.1 Why PBoC swap lines are the right laboratory . . . 3
2.2 PBoC swap-line data . . . 4
2.3 SWIFT payments data . . . 4
2.4 Outcome variables . . . 4
2.5 Sample selection . . . 5
2.6 Control variables . . . 5
2.7 RMB borrowing-cost data . . . 6
2.8 The 2015-16 RMB crisis as a mechanism test . . . 6
2.9 Heterogeneity variables . . . 6

3 Models and empirical specifications (all equations plus interpretation) **7**

3.1 Reduced-form jumpstart mechanism . . . 7
3.2 Baseline panel specification . . . 7
3.3 Intensive-margin outcomes . . . 7
3.4 Neighbor control . . . 7
3.5 Staggered-adoption estimator . . . 8
3.6 Borrowing-cost specification . . . 8
3.7 2015-16 crisis difference-in-differences . . . 8
3.8 Trade-finance heterogeneity test . . . 8
3.9 Currency-substitution regression . . . 8
3.10 Theoretical environment . . . 9
3.11 Imported inputs and production . . . 9
3.12 Borrowing costs . . . 9
3.13 Marginal cost . . . 9
3.14 Pricing-currency choices . . . 10

3.15 Profit and firm problem . . . 10
3.16 All-or-nothing finance choice . . . 11
3.17 Threshold for rising-currency credit . . . 11
3.18 Threshold for pricing in the rising currency . . . 11
3.19 How a swap line changes the borrowing-cost distribution . . . 11
3.20 Rising-currency pricing versus producer-currency pricing . . . 11
3.21 Model implication . . . 12

4 Identification and empirical design (what makes the estimates credible) 12
4.1 What identifies the main effect . . . 12
4.2 Why swap lines are plausibly relevant shocks . . . 12
4.3 Addressing China-integration confounds . . . 12
4.4 Addressing staggered-treatment bias . . . 13
4.5 Event-study timing and anticipation . . . 13
4.6 Spillovers . . . 13
4.7 2015-16 crisis as mechanism validation . . . 13
4.8 Trade-finance heterogeneity . . . 13
4.9 What the paper cannot fully prove . . . 14

5 Tables and figures 14
5.1 Figure 1: The PBoC swap lines . . . 14
5.2 Figure 2: RMB share in global payments and trade finance . . . 15
5.3 Figure 3: RMB payments per country versus trade with China . . . 16
5.4 Table 1: The PBoC's swap-line agreements as of end 2018 . . . 17
5.5 Figure 4: RMB payments before and after a swap line is signed . . . 18
5.6 Table 2: Swap lines and the probability that RMB is used . . . 19
5.7 Table 3: Swap lines and the intensive margin of RMB use . . . 20
5.8 Figure 5: Event-study plots . . . 21
5.9 Figure 6 and Table 4: RMB borrowing costs . . . 22
5.10 Figure 7 and Table 5: RMB payments during the 2015-16 RMB crisis . . . 23
5.11 Table 6: Swap lines, RMB trade finance, and country characteristics . . . 25
5.12 Table 7: Currency choice in payments with China . . . 26

6 Short summary 27

7 Conclusion 28
7.1 Core conclusion . . . 28
7.2 Economic intuition . . . 28
7.3 Takeaway . . . 28

1 Big picture

- **Question.** How can a currency that is not yet dominant become an international currency? The paper studies whether central bank policies can *jumpstart* the international use of a currency.
- **Policy shock.** The empirical object is the creation of RMB swap lines by the People's Bank of China (PBoC) between 2009 and 2018. These lines let foreign central banks borrow RMB and lend RMB liquidity to domestic banks.
- **Main mechanism.** A swap line reduces the right-tail risk of borrowing costs in the rising currency. Lower and more stable RMB funding costs make firms more willing to finance working capital in RMB. Because firms like to match the currency of costs and revenues under sticky prices, RMB working-capital finance complements RMB invoicing.
- **Data.** The paper combines PBoC swap-line data with monthly SWIFT data on cross-border payments by country pair, currency, and message type from October 2010 to October 2018.
- **Baseline sample.** The main sample is a balanced panel of 114 developing economies, with 21 countries treated by a new PBoC swap agreement during the sample period and 93 never-treated controls.
- **Main extensive-margin result.** Signing a swap line is associated with an increase of roughly 12-14 percentage points in the probability that a country uses RMB in cross-border payments.
- **Main intensive-margin result.** RMB payment values rise sharply after a swap line. Across specifications, the paper reports increases of roughly 220-450% in RMB use.
- **Timing.** Most of the extensive-margin effect appears within about 12 months of the agreement and then persists. Intensive-margin use grows over time.
- **Mechanism evidence.** Swap lines lower synthetic RMB borrowing costs, protect countries from the 2015-16 offshore RMB funding shock, raise RMB trade-finance messages, and produce larger effects in countries where the theory predicts stronger working-capital complementarities.
- **Currency substitution.** The RMB mainly replaces existing vehicle currencies such as the USD and EUR in payments with China, not the recipient country's home currency.
- **Economic takeaway.** International currency status is not only about trade size or financial openness. A lender-of-last-resort backstop for foreign trade finance can move a currency across a threshold from zero to positive international use.

2 Data and measurement (swap lines, RMB payments, and funding costs)

2.1 Why PBoC swap lines are the right laboratory

- A PBoC RMB swap line is an agreement with a foreign central bank. The foreign central bank can borrow RMB from the PBoC and supply RMB credit to local banks.
- The typical agreement is renewable and lasts three years. The line is an insurance device: even if it is rarely drawn, it caps extreme RMB funding costs.

- The PBoC's stated objective was to support RMB-denominated trade finance and settlement, so the policy maps closely to the paper's theoretical mechanism.
- The timing and cross-country pattern of swap-line agreements provide variation across countries and over time.

Interpretation. The swap lines are not ordinary trade agreements. They are lender-of-last-resort commitments in RMB. Their empirical relevance comes from changing the distribution of RMB borrowing costs faced by foreign banks and firms.

2.2 PBoC swap-line data

- The authors collect all PBoC swap-line agreements and renewals starting in 2009 from PBoC news releases and counterpart central-bank communications.
- By the end of 2018, 38 countries or monetary authorities had signed an RMB swap agreement.
- The treatment variable is
$$\text{SwapLine}_{i,t} = 1$$
if country i first signed a PBoC swap agreement at or before month t.
- The treatment is absorbing: once a country has signed a line, the indicator remains one even if the agreement temporarily lapses. The logic is that the potential for renewal preserves the insurance value.

Interpretation. The empirical design treats swap-line adoption as a staggered policy treatment. The policy is not measured by actual drawings; it is measured by access to the backstop.

2.3 SWIFT payments data

- The core outcome data come from SWIFT messages, aggregated monthly by country pair, currency, and message type.
- The sample period is October 2010 to October 2018.
- The main payment messages are MT103 and MT202, covering customer and bank-to-bank payment orders.
- The trade-finance and settlement messages are MT400 and MT700, covering advices of payment and letters of credit.
- The data measure the value of messages in USD. They exclude within-country messages.

Interpretation. SWIFT does not settle payments, but SWIFT messages are a broad proxy for cross-border payment activity and allow the authors to observe currency denomination across many countries.

2.4 Outcome variables

The paper defines the value of RMB payments sent or received by country i in month t as
$$\text{Rpayment}_{i,t}.$$

The extensive-margin outcome is

$$1(\text{Rpayment}_{i,t} > 0),$$

which equals one if the country uses RMB at all in that month.

The intensive-margin outcome is the RMB share of cross-border payments,

$$\text{Rshare}_{i,t},$$

scaled so that a unit change corresponds to 1 percentage point.

Interpretation. The paper separates a jump from zero use to positive use from subsequent growth in RMB payment shares. This is crucial because the theory is about jumpstarting international use, not only increasing already-large use.

2.5 Sample selection

- Large financial centers can generate misleading SWIFT observations because payments may be routed through several jurisdictions.
- The baseline sample therefore focuses on developing countries with average GDP per capita below 30,000 PPP dollars during the sample.
- Hong Kong and Macau are consolidated into China.
- Iran is excluded because sanctions make it a special case.
- The baseline also excludes countries that already had a swap line before the SWIFT sample begins, countries with average population below half a million, and countries missing control variables.
- The resulting baseline sample has 11,058 monthly observations on 114 countries: 21 treated countries and 93 controls.

Interpretation. The sample is designed to capture ordinary economies that may rely on foreign-currency trade finance, rather than payment hubs that intermediate other countries' transactions.

2.6 Control variables

The controls fall into three groups.

- **Trade integration with China:** exports and imports with China, trade shares relative to GDP, and a free-trade-agreement dummy.
- **Other China policies and political integration:** RMB clearing-bank presence, Asian Infrastructure Investment Bank membership, Chinese infrastructure investment flows, cumulative Chinese investment, and similarity of UN General Assembly voting with China.
- **Regional RMB trends:** the share of neighboring countries using RMB in the same month.

Interpretation. The central confound is that swap-line countries may also be countries becoming more integrated with China. These controls are intended to separate the swap-line association from broader China exposure.

2.7 RMB borrowing-cost data

- There is no comprehensive country-level dataset on RMB trade-finance interest rates.
- The paper instead constructs synthetic RMB borrowing costs using FX spot and swap markets.
- A local bank can synthetically borrow RMB by converting local currency into RMB and hedging the mismatch with an FX swap.
- The authors compute several synthetic borrowing routes and use the cheapest route as the relevant funding cost.
- The borrowing-cost panel covers 23 currencies from June 2007 to June 2021.

Interpretation. The synthetic rate proxies for the wholesale RMB funding cost available to local banks. If the swap line caps direct RMB borrowing, arbitrage should also cap synthetic RMB borrowing.

2.8 The 2015-16 RMB crisis as a mechanism test

- In August 2015, the PBoC adjusted the RMB's central parity rate, and the RMB depreciated.
- Because China maintained parallel onshore and offshore RMB markets, the PBoC drained offshore liquidity to defend the relationship between the two markets.
- This raised the level and volatility of offshore RMB borrowing costs.
- The model predicts that countries without swap lines should reduce RMB use, while countries with swap lines should be insulated.

Interpretation. This episode is a useful quasi-mechanism test because it shifts the private distribution of offshore RMB funding costs without being specific to a particular recipient country.

2.9 Heterogeneity variables

The paper tests whether swap lines matter more where the model predicts stronger complementarities:

- countries with above-median trade shares with China;
- countries that import more intermediate goods;
- countries whose export industries require more working capital.

Interpretation. If swap lines operate through trade finance and working capital, their effect should be stronger when RMB funding is especially useful for import inputs and export production.

3 Models and empirical specifications (all equations plus interpretation)

3.1 Reduced-form jumpstart mechanism

The empirical logic can be summarized as

$$\begin{aligned}\text{PBoC swap line}_{i,t} &\Rightarrow \text{lower RMB funding tail risk}_{i,t}\\ &\Rightarrow \text{more RMB trade finance}_{i,t}\\ &\Rightarrow \text{more RMB invoicing and payments}_{i,t}.\end{aligned}$$

Interpretation. The policy does not need to make RMB the dominant currency. It only needs to move some firms and countries from no RMB use to positive RMB use.

3.2 Baseline panel specification

The baseline regression is

$$1(\text{Rpayment}_{i,t} > 0) = \varsigma_i + \tau_t + \beta\,\text{SwapLine}_{i,t} + \gamma\,\text{Controls}_{i,t} + \text{error}_{i,t}. \tag{1}$$

- ς_i are country fixed effects.
- τ_t are month fixed effects.
- $\text{SwapLine}_{i,t}$ is the absorbing swap-line treatment.
- β measures the change in the probability of using RMB after a swap line is signed.

Interpretation. This is a difference-in-differences design with staggered adoption. It compares treated countries after adoption to their own pre-adoption behavior and to countries not yet treated or never treated.

3.3 Intensive-margin outcomes

The same specification is estimated with several intensive-margin outcomes:

$$\text{Rshare}_{i,t}, \qquad \ln(1 + \text{Rpayment}_{i,t}), \qquad \text{Rpayment}_{i,t}.$$

Interpretation. The share outcome gives a percentage-point change. The log-like outcome handles zeros while approximating proportional changes. The payment-value outcome is estimated with PPML to handle zeros and heteroskedasticity.

3.4 Neighbor control

The neighbor-use control is

$$\text{NeighborUse}_{i,t} = \frac{1}{|N_i|}\sum_{j\in N_i} 1(\text{Rpayment}_{j,t} > 0),$$

where N_i is the set of nearby countries.

Interpretation. This absorbs regional waves of RMB adoption that might otherwise be mistaken for the direct effect of country i's own swap line.

3.5 Staggered-adoption estimator

The paper mainly uses the imputation estimator of Borusyak, Jaravel, and Spiess (2024), with country-clustered standard errors and treatment effects averaged by cohort.

Interpretation. Standard two-way fixed effects can be biased when treatment effects vary over time. The imputation estimator addresses this problem in staggered difference-in-differences settings.

3.6 Borrowing-cost specification

The borrowing-cost test estimates the same type of staggered treatment effect using synthetic RMB borrowing costs as the outcome:

$$\text{SyntheticRMBRate}_{i,t} = \alpha_i + \delta_t + \beta\,\text{SwapLine}_{i,t} + u_{i,t}.$$

Interpretation. A negative β supports the model's key assumption: the swap line lowers or caps the borrowing costs of RMB outside China.

3.7 2015-16 crisis difference-in-differences

For the August 2015 episode, the paper compares countries that already had a swap line with those that did not:

$$\ln(1 + \text{Rpayment}_{i,t}) = \alpha_i + \delta_t + \beta\,(\text{SwapLineAug2015}_i \times \text{Post}_t) + X'_{i,t}\Gamma + u_{i,t}.$$

Interpretation. If the swap line insulates countries from offshore RMB funding spikes, then countries with swap lines should maintain RMB use after the shock relative to countries without swap lines.

3.8 Trade-finance heterogeneity test

For SWIFT trade-finance messages, the paper estimates the baseline model and then averages treatment effects separately for countries above and below the sample median in China trade share, intermediate-import intensity, and export working-capital needs.

Interpretation. The model predicts larger effects where RMB working-capital finance is more valuable. The heterogeneity tests check this prediction directly.

3.9 Currency-substitution regression

For payments with China, the paper uses the same treatment design with currency shares as outcomes:

$$\text{CurrencyShare}^c_{i,t} = \alpha_i + \delta_t + \beta_c\text{SwapLine}_{i,t} + u_{i,t},$$

where c is RMB, USD, EUR, GBP/JPY/CHF, other currencies, or the home currency.

Interpretation. The model predicts that RMB adoption should replace existing international vehicle currencies, not primarily local currencies.

3.10 Theoretical environment

The model has a continuum of firms indexed by $j \in [0,1]$. Each firm sells to many export markets $i \in [0,1]$, to the market of a dominant currency d, and to the market of a rising international currency r.

There are three periods:

- Period 0: the firm chooses the currency of imported inputs, trade finance, and sticky prices.
- Period 1: uncertainty about exchange rates, input costs, demand, and borrowing costs is resolved; the firm produces.
- Period 2: demand is served, revenues are collected, loans are repaid, and profits are realized.

Interpretation. The key novelty is that firms choose not only the currency of invoices, but also the currency of working-capital liabilities.

3.11 Imported inputs and production

The imported input aggregator is

$$x^j = \min \left\{ \frac{x_r^j}{\eta^j}, \frac{x_d^j}{1-\eta^j} \right\}. \tag{2}$$

The production function is

$$y^j = (x^j)^\alpha (l^j)^{1-\alpha}. \tag{3}$$

Here x^j is the imported input and l^j is the local input.

Interpretation. η^j governs the share of imported inputs and trade finance in the rising currency. The model assumes firms do not want currency mismatch between the input purchase and the financing of that input.

3.12 Borrowing costs

Borrowing b_d units of dominant-currency finance requires repayment of 1 unit of d. Borrowing b_r units of rising-currency finance requires repayment of ϵ^j units of r. Thus the interest rates are

$$\frac{1}{b_d} \quad \text{and} \quad \frac{\epsilon^j}{b_r}.$$

The random variable ϵ^j is drawn from $G^j(\epsilon^j)$.

Interpretation. The dominant currency is attractive because its funding cost is stable and liquid. The rising currency has potentially high right-tail borrowing-cost risk.

3.13 Marginal cost

The firm's marginal cost is

$$C(\eta^j, \epsilon^j, s_r, s_d, w) = \left[\frac{\eta^j s_r \rho_r \left(\epsilon^j / b_r\right) + (1-\eta^j) s_d \rho_d \left(1/b_d\right)}{\alpha} \right]^\alpha \left(\frac{w}{1-\alpha} \right)^{1-\alpha}. \tag{4}$$

Interpretation. Marginal cost depends on the currency of imported inputs and finance, exchange rates, input prices, and local production costs. If the firm borrows and buys inputs in RMB, RMB exchange-rate movements enter costs directly.

3.14 Pricing-currency choices

In each market i, the firm can choose producer-currency pricing, local-currency pricing, dominant-currency pricing, or rising-currency pricing:

$$P_i^j \in \{PCP, LCP, DCP, RCP\}.$$

Demand under local-currency pricing is

$$y_i^j = \left(\frac{p_i^j}{q_i}\right)^{-\theta}.$$

Under producer-currency pricing,

$$y_i^j = \left(\frac{p_i^j}{q_i s_i}\right)^{-\theta}.$$

Under dominant-currency pricing,

$$y_i^j = \left(\frac{p_i^j s_d}{q_i s_i}\right)^{-\theta},$$

and under rising-currency pricing,

$$y_i^j = \left(\frac{p_i^j s_r}{q_i s_i}\right)^{-\theta}.$$

Interpretation. Sticky nominal prices make currency choice matter. A firm wants the currency of revenues to co-move with the currency of costs.

3.15 Profit and firm problem

For example, under local-currency pricing, ex-post profit is

$$\pi^{LCP}(p_i^j, \eta^j, \epsilon^j, S, Q, w) = (p_i^j s_i)\left(\frac{p_i^j}{q_i}\right)^{-\theta} - C(\eta^j, \epsilon^j, S, w)\left(\frac{p_i^j}{q_i}\right)^{-\theta}. \tag{5}$$

The firm's problem can be written schematically as

$$\max_{\eta^j}\left(\int_0^1 \max_{P_i^j}\max_{p_i^j}\left[\int\int \pi^P(p_i^j, \eta^j, \epsilon^j, S, w)\, dH(S, Q, w)\, dG^j(\epsilon^j)\right] di + \cdots\right). \tag{6}$$

Interpretation. The firm chooses input-finance currency at the firm level and pricing currency at the market level. The omitted terms represent the positive-mass dominant-currency and rising-currency markets.

3.16 All-or-nothing finance choice

One model result is

$$\eta^j \in \{0, 1\}.$$

Interpretation. Firms do not diversify finance currencies in the benchmark model. One currency weakly dominates once expected cost and hedging benefits are considered.

3.17 Threshold for rising-currency credit

A firm chooses rising-currency credit if

$$\left(\int (\epsilon^j)^\alpha dG^j(\epsilon^j) \right)^{1/\alpha} \leq \left(\frac{b_r}{b_d} \right) \left(\frac{\rho_d}{\rho_r} \right) \Psi(\mu, \Sigma, P^j). \tag{7}$$

Interpretation. The left side summarizes expected rising-currency borrowing costs. The right side captures relative funding costs, input costs, exchange-rate hedging, and pricing complementarities. A swap line can push firms across this threshold.

3.18 Threshold for pricing in the rising currency

If the firm uses rising-currency credit, rising-currency pricing is preferred to local-currency pricing in market i when

$$\sigma_i^2 - 2\alpha\sigma_{ir} - 2(1-\alpha)\sigma_{iw} \geq \Phi, \qquad \Phi = \sigma_r^2 - 2\alpha\sigma_r^2 - 2(1-\alpha)\sigma_{rw}. \tag{8}$$

Interpretation. The more volatile the local exchange rate is relative to the rising currency, and the better the rising currency hedges costs, the more attractive rising-currency invoicing becomes.

3.19 How a swap line changes the borrowing-cost distribution

The swap line is modeled as an option to borrow rising-currency funds at a known rate ϵ^{swap}/b_r. It truncates the effective distribution of borrowing costs:

$$\widetilde{G}^j(\epsilon^j) = \begin{cases} G^j(\epsilon^j)/G^j(\epsilon^{swap}), & \epsilon^j < \epsilon^{swap}, \\ 1, & \epsilon^j \geq \epsilon^{swap}. \end{cases} \tag{9}$$

Interpretation. The swap line cuts off extreme high borrowing costs. It may be rarely used ex post, but it changes ex-ante choices because firms and banks know the tail is capped.

3.20 Rising-currency pricing versus producer-currency pricing

The model also gives a condition under which rising-currency pricing beats producer-currency pricing:

$$\sigma_{rw} \geq \Omega, \qquad \Omega = \sigma_r^2 \left(\frac{0.5 - \alpha}{1 - \alpha} \right). \tag{10}$$

Interpretation. If the rising currency co-moves enough with noncredit marginal costs, or if imported inputs are important enough, firms prefer to price in the rising currency rather than in their own currency.

3.21 Model implication

Most currencies fail to become international because they do not clear the relevant thresholds: exchange rates are too volatile, funding costs are too risky, capital markets are too shallow, or the issuing country is too small in trade. The RMB was closer to the thresholds because China was a large exporter, the RMB was relatively stable against the USD during the sample, and PBoC policy lowered foreign RMB funding risk.

Interpretation. The theory explains why the policy worked for the RMB but would not necessarily work for every currency.

4 Identification and empirical design (what makes the estimates credible)

4.1 What identifies the main effect

- The core identifying variation is country-by-month variation in swap-line adoption.
- Treated countries are compared before and after adoption, while month fixed effects absorb global trends in RMB use.
- Country fixed effects absorb persistent country differences in financial sophistication, China exposure, geography, and institutions.

Interpretation. The main coefficient is credible if, conditional on controls, countries that signed swap lines would otherwise have followed similar RMB-use trends as control countries.

4.2 Why swap lines are plausibly relevant shocks

- The PBoC stated that the lines were intended to support RMB trade finance and settlement.
- The lines act as insurance against extreme RMB borrowing costs.
- The paper directly shows that synthetic RMB borrowing costs fall after a swap line.

Interpretation. The mechanism is not inferred only from payment outcomes. The authors also test the intermediate object: RMB funding costs.

4.3 Addressing China-integration confounds

- The baseline includes controls for trade with China.
- It also controls for other China-related policies, such as RMB clearing banks, AIIB membership, and Chinese infrastructure investment.
- Appendix exercises show that the results are not explained by Belt and Road membership and appear even for payments not directly involving China.

Interpretation. These checks reduce the concern that swap lines merely proxy for rising trade or political integration with China.

4.4 Addressing staggered-treatment bias

- The paper uses the Borusyak-Jaravel-Spiess imputation estimator for staggered difference-in-differences.
- It also reports standard two-way fixed-effects and synthetic difference-in-differences robustness checks.

Interpretation. This matters because treatment effects may grow over time, and ordinary two-way fixed effects can compare already-treated units to later-treated units in problematic ways.

4.5 Event-study timing and anticipation

- The extensive-margin event study shows little evidence of long pre-trends, although there is movement shortly before some agreements.
- The authors interpret this short pre-effect as possible anticipation because negotiations or announcements may precede formal signing.
- The intensive-margin event study shows no clear pre-trend and treatment effects that grow after adoption.

Interpretation. Anticipation does not necessarily undermine the mechanism because the insurance value can affect behavior once the agreement becomes expected.

4.6 Spillovers

- If one country starts invoicing and paying in RMB, nearby trading partners may also start using RMB.
- Such spillovers are evidence that the swap line affects international currency use, but they complicate a strict stable-unit-treatment-value assumption.
- The paper explicitly studies spillovers to neighboring countries in appendix exercises.

Interpretation. Spillovers strengthen the economic story but mean the direct treatment coefficient may understate broader network effects.

4.7 2015-16 crisis as mechanism validation

- The offshore RMB funding shock raised private borrowing-cost volatility.
- Countries without swap lines reduced RMB use sharply.
- Countries with swap lines were protected and maintained RMB use.

Interpretation. This is one of the strongest pieces of evidence because it tests the model's exact claim: the distribution of RMB borrowing costs affects RMB adoption.

4.8 Trade-finance heterogeneity

- Swap-line effects are stronger in countries with larger trade shares with China.

- They are stronger in countries importing more intermediate inputs.
- They are stronger in countries whose exports are in industries with greater working-capital needs.

Interpretation. These patterns are difficult to explain with a generic RMB internationalization story alone; they fit the paper's working-capital mechanism.

4.9 What the paper cannot fully prove

- Swap lines are not randomly assigned.
- Political negotiations and China-related policy packages may influence both swap-line adoption and RMB use.
- SWIFT data are message data, not complete settlement data.
- Some payment flows may be routed through financial centers, creating measurement concerns.
- The analysis documents strong associations and mechanism-consistent patterns, but the authors are careful not to claim a perfectly watertight causal design.

Interpretation. The most persuasive evidence is the full pattern: reduced borrowing costs, extensive-margin RMB adoption, persistent intensive growth, 2015-16 insulation, trade-finance effects, and predicted heterogeneity.

5 Tables and figures

5.1 Figure 1: The PBoC swap lines

Read:

- Panel (a) shows rapid growth in the number of PBoC swap-line agreements and in the notional amount of active lines.
- Most growth occurs in the first half of the 2010s, followed by a slowdown after 2016.
- Panel (b) maps active lines in 2018 and shows broad geographic coverage.
- Large financial centers have large lines, but many nonfinancial-center economies also sign agreements.

Economic message. The PBoC created a global RMB liquidity network before the RMB was a dominant international currency. This policy network is the empirical variation behind the paper.

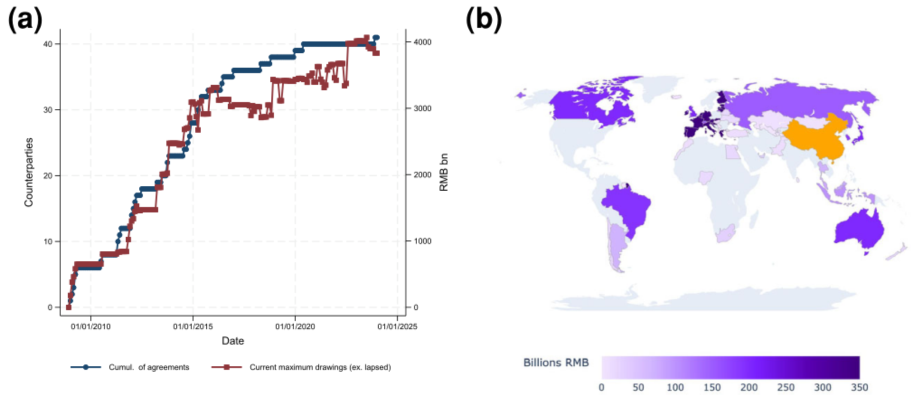

FIGURE 1

The PBoC swap lines: (a) swap lines: number and amounts and (b) the network of swap lines in 2018

Notes: In Panel (a), the navy line with circular dots shows the cumulative number of countries/central banks that had signed at least one swap agreement with the PBoC; the burgundy line with square dots shows the current notional limit on all active PBoC swap lines where lines that have lapsed and have not been renewed receive a zero value. Panel (b) shows swap lines active at the end of 2018; the shade of color indicates the maximum amount the PBOC is willing to lend in RMB bn.

5.2 Figure 2: RMB share in global payments and trade finance

Read:

- Panel (a) shows the RMB share of global SWIFT payments using MT103 and MT202 messages.
- Panel (b) shows the RMB share in trade-finance and settlement messages using MT400 and MT700.
- RMB use rises after the early 2010s, stalls around 2015-16, and then resumes growth.
- The 2015-16 stall is especially visible for trade finance.

Economic message. The RMB's internationalization is visible in aggregate payments, but the time-series pattern alone cannot identify the effect of swap lines. The paper therefore turns to cross-country variation.

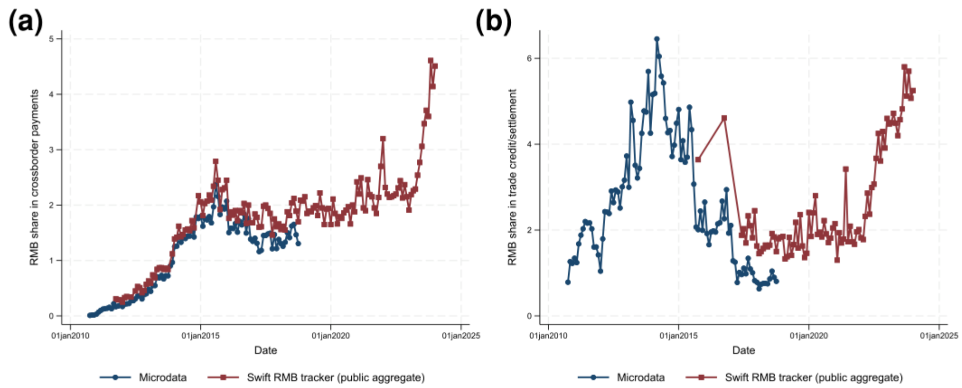

FIGURE 2
RMB share in global payments and trade finance: (a) payments (MT103 & MT202) and (b) trade finance/settlement (MT400 & MT700)

Notes: In Panel (a), the burgundy line with square dots shows the percentage of SWIFT messages MT103 and MT202 denominated in RMB as reported by the SWIFT RMB tracker; the navy line with circular dots shows the equivalent for our microdata. Panel (b) is the equivalent but for message types MT400 and MT700, i.e. those related to trade finance. The navy and burgundy lines do not align precisely due to differences in how jurisdictions have been consolidated.

5.3 Figure 3: RMB payments per country versus trade with China

Read:

- The figure plots each country's average RMB payment share against its goods-trade share with China.
- Many countries trade substantially with China but use little or no RMB.
- Heavy RMB users include special cases such as financial centers and economies with unusually close China links.
- The RMB generally punches below China's weight in trade, unlike the USD.

Economic message. Trade exposure alone does not explain RMB use. This motivates a theory in which funding-cost stability and currency complementarities matter.

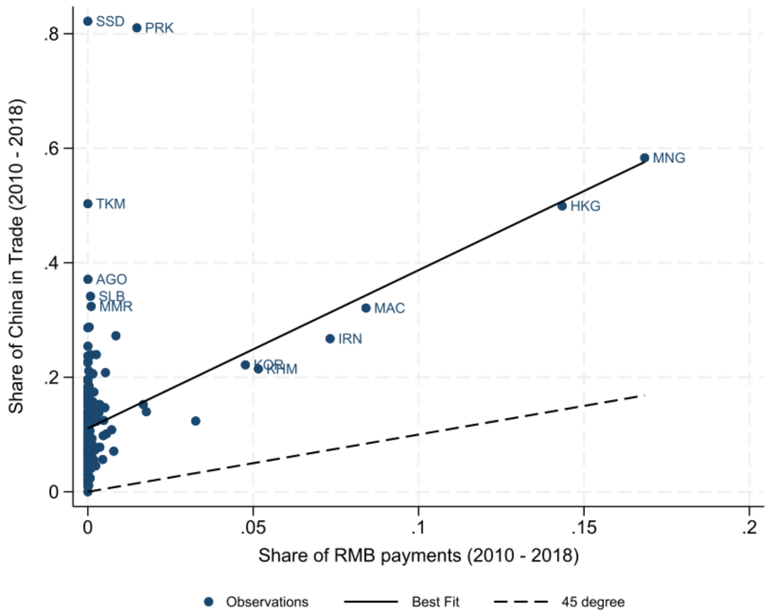

FIGURE 3
RMB payments per country versus trade with China
r plot showing the average share of a country's good trade with China (sum of imports and exports) on the y-axis an
are of payments in RMB (MT103 and MT202) on the *x*-axis. The dashed line is a 45 degree line and the solid black li
fit. Data on trade shares are from the IMF direction of trade statistics.

5.4 Table 1: The PBoC's swap-line agreements as of end 2018

Read:

- The table lists every PBoC swap-line agreement through 2018, the date of first agreement, and the initial notional amount.
- Some countries enter the baseline sample; others are excluded because they signed before the sample start or exceed the income threshold.
- Examples of baseline treated countries include Albania, Brazil, Chile, Egypt, Hungary, Kazakhstan, Mongolia, Morocco, Nigeria, Pakistan, Russia, South Africa, Thailand, Turkey, and Ukraine.
- Notional amounts range from small lines such as RMB 1 billion to very large lines such as RMB 350 billion for the ECB.

Economic message. The swap-line network spans countries that differ widely in geography, size, and China exposure. This creates useful cross-country variation.

TABLE 1

The PBoC's swap line agreements as of end 2018

Country	In Baseline Sample	Date of First Agreement	Notional Amount (RMB mn) as of First Agreement	Country	In Baseline Sample	Date of First Agreement	Notional Amount (RMB mn) as of First Agreement
Albania	✓	12/09/2013	2,000	Malaysia		08/02/2009	80,000
Argentina		02/04/2009	70,000	Mongolia	✓	06/05/2011	5,000
Armenia	✓	25/03/2015	1,000	Morocco	✓	11/05/2016	10,000
Australia		22/03/2012	200,000	New Zealand		18/04/2011	25,000
Belarus		11/03/2009	20,000	Nigeria	✓	27/04/2018	15,000
Brazil	✓	26/03/2013	190,000	Pakistan	✓	23/12/2011	10,000
Canada		08/11/2014	200,000	Qatar		03/11/2014	35,000
Chile	✓	25/05/2015	22,000	Russia	✓	13/10/2014	150,000
ECB		08/10/2013	350,000	Serbia	✓	17/06/2016	1,500
Egypt	✓	06/12/2016	18,000	Singapore		23/07/2010	150,000
Hong Kong		20/01/2009	200,000	South Africa	✓	10/04/2015	30,000
Hungary	✓	09/09/2013	10,000	Sri Lanka	✓	16/09/2014	10,000
Iceland		09/06/2010	3,500	Surinam	✓	18/03/2015	1,000
Indonesia		23/03/2009	100,000	Switzerland		21/07/2014	150,000
Japan		26/10/2018	200,000	Tajikistan	✓	03/09/2015	3,000
Kazakhstan	✓	13/06/2011	7,000	Thailand	✓	22/12/2011	70,000
Korea		20/04/2009	180,000	Turkey	✓	21/02/2012	10,000
Malaysia		08/02/2009	80,000	Ukraine	✓	26/06/2012	15,000
Mongolia	✓	06/05/2011	5,000	UAE		17/01/2012	35,000

Notes: This shows all the countries that signed swap agreements with the PBoC until the end of our sample in 2018. The second column indicates whether the country enters our baseline regression sample (*i.e.* column (4) of Table 2); countries can be excluded if they signed an agreement before the start of our sample in October 2010 or are above the per capita income threshold. Column (10) of Table 2 relaxes this income threshold filter for the main empirical analysis.

5.5 Figure 4: RMB payments before and after a swap line is signed

Read:

- Panels (a) and (b) show that treated countries use little RMB before signing a swap line and more RMB afterward.
- The control group of never-treated countries remains near zero in panel (b).
- Panels (c) and (d) show country-level before-after averages; most treated countries move above the 45-degree line.
- Mongolia is an outlier for intensive RMB use and is excluded from some intensive-margin specifications.

Economic message. The raw event-time evidence already suggests a jumpstart: RMB use rises after swap-line adoption.

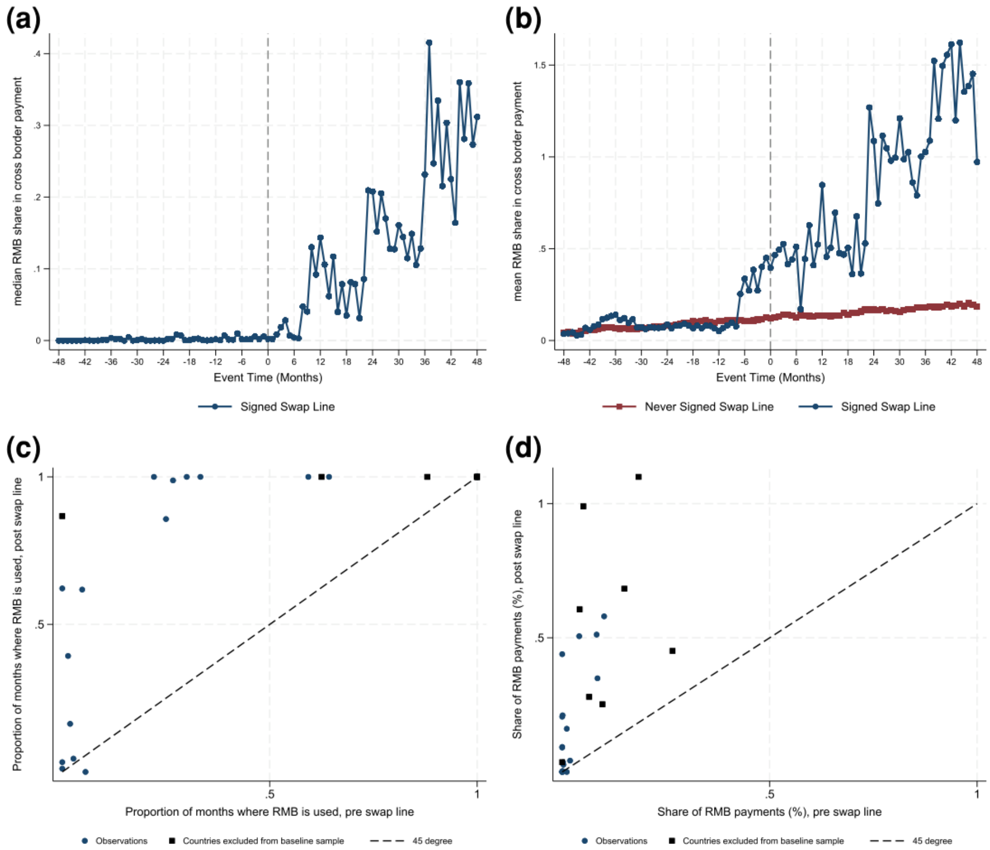

FIGURE 4

RMB payments before and after a swap line is signed: (a) median of $\text{Rshare}_{i,t}$, (b) mean of $\text{Rshare}_{i,t}$, (c) country means of $1(\text{Rpayment}_{i,t} > 0)$, and (d) country means of $\text{Rshare}_{i,t}$

Notes: Panels (a) and (b) plot of $\text{Rshare}_{i,t}$ against event time. Event time is defined such that month zero corresponds to the month when the country first signs a PBoC swap line. Panel (a) plots, for each event time period, the median value of $\text{Rshare}_{i,t}$ for all countries that have signed a swap agreement in our sample. The navy line with circular dots in Panel (b) presents the equivalent value for the mean of countries that have signed swap line agreements. The burgundy line with square dots in Panel (b) corresponds to a control group based on countries that have never signed swap agreements. To produce it, for each country that signed an agreement, we take a mean of $\text{Rshare}_{i,t}$ *for the countries who never entered an agreement in the same event time period. This forms a control group for each country that entered an agreement. We then take a second mean of these control series across the swap line countries for each event time period. This second mean is the burgundy line. The median RMB usage for countries that have not signed a swap line is nil for all time periods so we do not present an equivalent series for Panel (a). Panel (c) plots, for each country that has signed an agreement, the average level of* $1(\text{Rpayment}_{i,t} > 0)$ before and after signing a swap line. Navy circles indicate countries that enter our baseline sample, and burgundy squares indicate developed economies and financial centers that we drop in the baseline specification. Panel (d) plots the equivalent for $\text{Rshare}_{i,t}$, Mongolia is excluded.

5.6 Table 2: Swap lines and the probability that RMB is used

Read:

- Column (1), with only country and time fixed effects, estimates an 0.1065 increase in the probability of RMB use.
- Column (4), with trade, policy, and neighbor controls, estimates 0.1179.
- Payments sent respond more strongly than payments received: column (6) is 0.1168, while column

(7) is 0.0551 and not statistically significant.

- Shifting treatment back six months raises the estimate to 0.1780, consistent with anticipation or early implementation.
- Excluding Mongolia gives 0.1479; including developed economies gives 0.0983.

Economic message. The extensive-margin result is robust: signing a swap line is associated with a materially higher probability that the country uses RMB at all.

TABLE 2
Swap lines and the prob. the RMB is used

	Baseline Specification				Robustness					
	Time & Country f.e. (1)	Incl. China Trade (2)	Incl. China Policy (3)	Incl. Neigh Share (4)	Least Squares (5)	Payments Sent (6)	Payments Rec'd (7)	6 Month Shift (8)	Ex Mongolia (9)	Incl. Developed (10)
$\text{SwapLine}_{i,t}$	0.1065^{**}	0.1069^{**}	0.1229^{***}	0.1179^{***}	0.1548^{**}	0.1168^{**}	0.0551	0.1780^{***}	0.1479^{***}	0.0983^{**}
	(0.046)	(0.045)	(0.044)	(0.046)	(0.071)	(0.046)	(0.039)	(0.048)	(0.047)	(0.049)
N treated	21	21	21	21	21	21	21	21	20	29
N control	93	93	93	93	93	93	93	93	93	107
T	97	97	97	97	97	97	97	97	97	97
Time f.e.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country f.e.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Trade Controls	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Policy Controls	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Neighbor Control	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Method	BJS(24)	BJS(24)	BJS(24)	BJS(24)	OLS	BJS(24)	BJS(24)	BJS(24)	BJS(24)	BJS(24)

Notes: Standard errors clustered by country in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
Estimates of equation (3.1). In all specifications, the outcome variable is an indicator variable for whether the country makes a payment denominated in RMB in a particular month where payment is defined by SWIFT message types MT 103 and MT 202. The treatment variable is a dummy variable indicating whether the country's central bank, as of month t, has ever signed a swap line agreement with the PBoC ($\text{SwapLine}_{i,t}$). Sample period is October 2010 to October 2018. BJS(24) refers to the DiD imputation method from Borusyak *et al.* (2024) . Column (1): includes only country and time fixed effects and no further controls. Column (2): as previous, as but includes as controls a Chinese FTA dummy and trade flows with China. Column (3), as previous, but includes as extra controls dummies for membership of the AIIB and the presence of an RMB clearing bank and Chinese investment flows into the country. Column (4): as previous, but includes Neighbor $\text{Use}_{i,t}$ as an extra control. Column (5): as column (4), but uses a two-way fixed effects estimator. Column (6): as column (4), but only considers payments sent. Column (7): as column (4), but only considers payments received. Column (8): as column (4), but shifts treatment back by six months. Column (9): excludes Mongolia as an outlier. Column (10): as column (4), but relaxes sample selection criteria based on country GDP per capita.

5.7 Table 3: Swap lines and the intensive margin of RMB use

Read:

- Using RMB payment share, the baseline coefficient is 0.1289 without controls and 0.0948 with controls.
- Treatment effects compound over time: 0.0262 in months 0-11, 0.0772 in months 12-23, 0.1119 in months 24-35, and 0.2983 in months 36-48.
- Using ln(1+Rpayment), estimates remain positive and significant across imputation and synthetic DiD specifications.
- PPML estimates on payment values are also positive and significant.

Economic message. The swap line not only induces first-time RMB use. It is also associated with growing RMB payment volumes over subsequent years.

TABLE 3

Swaplines and the intensive margin of RMB use

Outcome Variable:	$\text{Rshare}_{i,t}$					$ln(1+\text{Rpayment}_{i,t})$				$\text{Rpayment}_{i,t}$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
$\text{SwapLine}_{i,t}$	0.1289^{***}	0.0948^{***}		0.1106^{**}	0.1008^{**}	3.7154^{***}	3.4016^{***}	3.2354^{**}	2.5637^{*}	1.6831^{***}	1.1226^{***}
	(0.019)	(0.011)		(0.043)	(0.045)	(0.720)	(0.773)	(1.482)	(1.436)	(0.298)	(0.306)
0–11 months			0.0262^{***}								
			(0.009)								
12–23 months			0.0772^{***}								
			(0.007)								
24–35 months			0.1119^{**}								
			(0.048)								
36–48 months			0.2983^{***}								
			(0.014)								
N treated	20	20	20	20	20	20	20	20	20	20	20
N control	93	93	93	93	93	93	93	93	93	58	58
T	97	97	97	97	97	97	97	97	97	97	97
Time f.e.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country f.e.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Trade Controls	No	Yes	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Policy Controls	No	Yes	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Neighbor Control	No	Yes	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Log Controls	No	No	No	No	No	No	Yes	No	Yes	No	Yes
Method	BJS(24)	BJS(24)	BJS(24)	SDID	SDID	BJS(24)	BJS(24)	SDID	SDID	PPML	PPML

Notes: Standard errors clustered by country in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
Estimates of equation (3.1) with different outcome variables reflecting the intensive margin of RMB use, *i.e.* the value of RMB payments in a particular month where payment is defined by SWIFT message types MT 103 and MT 202. The treatment variable is a dummy variable indicating whether the country's central bank, as of month t, has ever signed a swap line agreement with the PBoC ($\text{SwapLine}_{i,t}$). Sample period is October 2010 to October 2018, Mongolia is excluded from the set of treated countries. BJS(24) refers to the DiD imputation method from Borusyak *et al.* (2024). SDID corresponds to a synthetic difference in differences estimator from Arkhangelsky *et al.* (2021), in which case bootstrapped standard errors are reported. PPML corresponds to a psuedo Poisson maximum likelihood estimator (Santos Silva and Tenreyro, 2006). Trade Controls are a Chinese FTA dummy and goods imports and exports from/to China in log terms and as a share of the country total. Policy Controls are dummies for membership of the AIIB and the presence of an RMB clearing bank and variables capturing Chinese investment flows into the country. Neighbor control captures the share of neighboring countries using the RMB. Log Controls are the logarithm of the country's GDP and the logarithm of its total cross border payments, both in USD. Column (1): $Rshare_{i,t}$ is the outcome variable, specification includes only country and time fixed effects and no further controls. Column (2): as previous, but includes all controls. Column (3) as previous, but shows treatment effects after years 1, 2, and 3. Column (4) and (5): as columns (1) and

5.8 Figure 5: Event-study plots

Read:

- Panel (a) plots the treatment effect for the extensive-margin outcome.
- Most of the extensive-margin effect appears near the signing date and stabilizes after about 12 months.
- Panel (b) plots the treatment effect for RMB payment share.
- The intensive-margin event study shows little evidence of pre-trends and a persistent post-treatment increase.

Economic message. The timing supports a policy interpretation: RMB use rises after swap-line adoption and does not quickly revert.

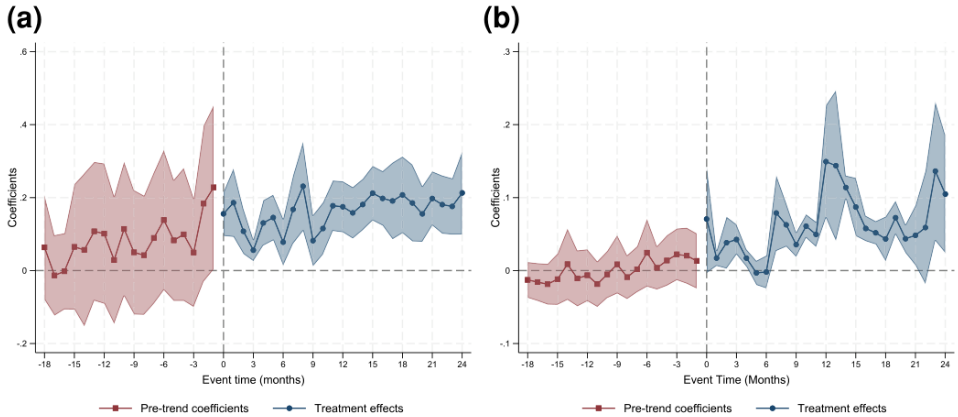

FIGURE 5
Event study plots: (a) $1(\text{Rpayment}_{i,t} > 0)$ and (b) $\text{Rshare}_{i,t}$

Notes: Event study plots using the methodology of Borusyak *et al.* (2024). Panel (a) presents event study plots between horizons $-18/+24$ months for the specification in column (4) of Table 2. Panel (b) presents the equivalent for column (2) of Table 3. The shaded areas represent 95% confidence intervals.

5.9 Figure 6 and Table 4: RMB borrowing costs

Read:

- Figure 6(a) shows onshore RMB rates, offshore RMB rates, and average synthetic RMB borrowing costs.
- Offshore RMB borrowing costs spike during the 2015-16 RMB episode.
- Figure 6(b) shows that synthetic RMB borrowing costs fall after a swap line is signed.
- Table 4 estimates a baseline borrowing-cost effect of -1.1539 percentage points.
- The effect is larger for emerging markets only: -2.0505 percentage points.

Economic message. This is direct evidence for the model's first step: swap lines reduce RMB funding-cost risk outside China.

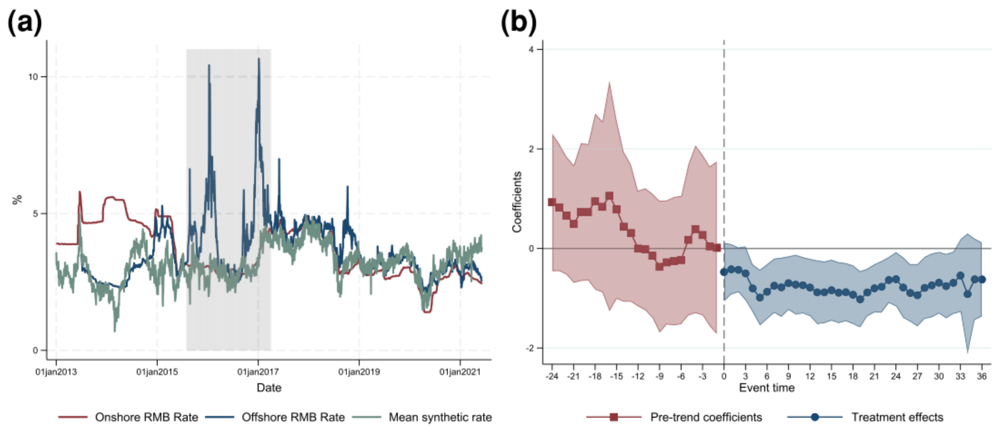

FIGURE 6

RMB borrowing costs: (a) time series of borrowing costs and (b) event study around swap agreements

Notes: Panel (a) presents time series plots of the 3-month RMB SHIBOR rate, the 3-month HIBOR rate, and the average of the synthetic RMB borrowing costs we compute for countries in our sample as discussed in Appendix B.1. Panel (b) is an event study plots using the methodology of Borusyak *et al.* (2024) based on the equivalent specification to column (1) of Table 4 with observations aggregated to a monthly frequency by taking averages. Shaded areas represent 95% confidence intervals.

TABLE 4

Swap lines and RMB borrowing costs

	Baseline (1)	Least Squares (2)	Spread v China Rate (3)	1-year Tenor (4)	Emerging Markets Only (5)
$\text{SwapLine}_{i,t}$	-1.1539^{***}	-0.4953^{*}	-1.1967^{***}	-0.9415^{***}	-2.0505^{*}
	(0.425)	(0.288)	(0.425)	(0.321)	(1.090)
N	23	23	23	23	13
T (trading days)	3,506	3,506	3,506	3,506	3,506
Time f.e.	Yes	Yes	Yes	Yes	Yes
Country f.e.	Yes	Yes	Yes	Yes	Yes
Method	BJS(24)	OLS	BJS(24)	BJS(24)	BJS(24)

Notes: Standard errors clustered by country in parentheses * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
Sample covers twenty-three currencies in a balanced panel covering trading days from 1 June 2007 to 8 June 2021. The outcome variable is the country-specific estimate of the synthetic RMB borrowing cost as computed in Appendix B. The treatment variable is a dummy variable indicating whether the country's central bank, as of trading day t, has ever signed a swap line agreement with the PBoC. Column (1): baseline specification estimated using the imputation method of Borusyak *et al.* (2024). Column (2): uses a two-way fixed effects estimator rather than a the imputation method. Column (3): redefines the the outcome variable to be the spread over the equivalent offshore or onshore Chinese borrowing cost. Column (4): uses a 1-year tenor rather than a 3-month tenor. Column (5): restricts the sample only to the emerging market countries used in the main analysis sample reduced to thirteen currencies.

5.10 Figure 7 and Table 5: RMB payments during the 2015-16 RMB crisis

Read:

- Figure 7 compares RMB use by countries with and without a swap line as of August 2015.
- Countries without swap lines experience a sharp fall in RMB use after the offshore RMB funding shock.

- Countries with swap lines maintain high RMB use.
- Table 5 estimates positive post-crisis differences for swap-line countries; the baseline coefficient is 2.2141 log points and rises to 2.7522 with controls.
- Synthetic DiD specifications give similar estimates.

Economic message. Swap lines act like insurance. When private offshore RMB borrowing costs become volatile, countries with access to PBoC liquidity keep using RMB.

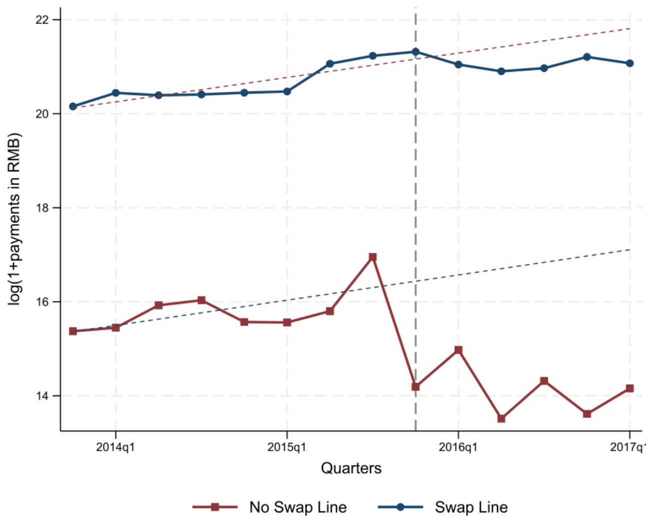

FIGURE 7
RMB payments before and after the 2015–2016 RMB crisis

Notes: The figure plots $ln(1 + \text{Rpayment}_{i,t})$ for countries with and without swap agreements as of August 2015. $\text{Rpayment}_{i,t}$ has been aggregated to a quarterly frequency. To be included in the sample a country must make positive RMB payments in all quarters between 2013-Q4 and 2015-Q3 as well as meet our standard sample selection criteria. Dashed lines show linear trend lines computed over 2013-Q4 and 2015-Q3. Lines colored in navy with circular dots relate to countries with a swap agreement, lines burgundy with square dots to those without.

TABLE 5

Swap lines and RMB payments during the August 2015 episode

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Swap Line Aug-15$_i$	2.2141*	2.7522**	1.7645*	2.2315*	2.5698**	2.9280**	1.9584*	2.1724
$\times$Post$_t$	(1.181)	(1.307)	(1.068)	(1.200)	(1.203)	(1.347)	(1.095)	(1.370)
N treated	12	12	12	12	12	12	12	12
N control	17	17	17	17	17	17	17	17
T	14	14	14	14	14	14	14	14
Time f.e.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country f.e.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes	No	Yes	No	Yes
Event Date	2015Q3	2015Q3	2015Q2	2015Q2	2015Q3	2015Q3	2015Q2	2015Q2
Method	OLS	OLS	OLS	OLS	SDID	SDID	SDID	SDID

Notes: Standard errors clustered by country in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Difference-in-differences estimates based on an outcome variable of $ln(1 + \text{Rpayment}_{i,t})$, where $\text{Rpayment}_{i,t}$ is the total value of payments in RMB made by country i in quarter t. The treatment variable is a dummy that takes a value of one if the country has a PBoC swap line in August 2015. The sample period is 2013:Q4–2017:Q1. The sample applies the same selection criteria as in the main analysis; in addition, all included countries must have made use of the RMB in every quarter between 2013:Q4 and 2015:Q2. Column (1) presents results with country and time fixed effects and event date of 2015 Q3; Column (2) adds controls including the logarithm of the country's overall payments, its nominal GDP, and its trade with China. Column (3) and (4) repeat prior two columns with an event date of 2015 Q2. Columns (5)–(8) repeat the prior estimation using a synthetic differences-in-differences estimator (Arkhangelsky *et al.*, 2021).

5.11 Table 6: Swap lines, RMB trade finance, and country characteristics

Read:

- For trade-finance messages, the extensive-margin effect is 0.1496 without controls and 0.1262 with controls.
- The RMB share of trade-finance messages rises by 0.1657 percentage points.
- Countries with above-median trade shares with China have a much larger effect: 0.1804 versus 0.0234 below median.
- Countries with above-median intermediate-import shares have a larger effect: 0.1670 versus 0.0523.
- Countries with above-median export working-capital needs have a larger effect: 0.1445 versus 0.0772.

Economic message. The swap-line effect is strongest where RMB working-capital finance should matter most. This supports the paper's trade-finance mechanism.

TABLE 6

Swaplines, RMB trade finance, and country characteristics

				China Trade Share		Intermediate Imports		Export Working Capital Needs	
	$1(\text{Rtradecredit}_{i,t} > 0)$		$\text{RtradecreditShare}_{i,t}$	below	above	below	above	below	above
				median	median	median	median	median	median
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$\text{SwapLine}_{i,t}$	0.1496***	0.1262***	0.1657***	0.0234	0.1804***	0.0523**	0.1670***	0.0772***	0.1445***
	(0.009)	(0.022)	(0.005)	(0.016)	(0.025)	(0.024)	(0.021)	(0.023)	(0.022)
Difference				0.1571***		0.1147***		0.0673***	
in coefficients				(0.010)		(0.006)		(0.005)	
N treated	21	21	21	21		21		21	
N control	93	93	93	93		93		93	
T	97	97	97	97		97		97	
Time f.e.	Yes	Yes	Yes	Yes		Yes		Yes	
Country f.e.	Yes	Yes	Yes	Yes		Yes		Yes	
Trade Controls	No	Yes	Yes	Yes		Yes		Yes	
Policy Controls	No	Yes	Yes	Yes		Yes		Yes	
Neighbor Control	No	Yes	Yes	Yes		Yes		Yes	
Method	BJS(24)	BJS(24)	BJS(24)	BJS(24)		BJS(24)		BJS(24)	

Notes: Standard errors clustered by country in parentheses, * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
Estimates of equation (3.1) with different outcome variables reflecting the intensive margin of RMB use, *i.e.* the value of RMB payments in a particular month. In this table, payments are defined as messages related to trade finance and/or settlement, that is message types MT400 and MT700 in SWIFT. The treatment variable is a dummy variable indicating whether the country's central bank, as of month *t*, has ever signed a swap line agreement with the PBoC ($\text{SwapLine}_{i,t}$). Sample period is October 2010 to October 2018. BJS(24) refers to the did imputation method from Borusyak *et al.* (2024). Columns (1) and (2) consider the extensive margin effect with and without the control variables defined in the main text. Column (3) repeats the exercise with the share of messages related to trade finance that are denominated in RMB as the outcome variable. If the country reports no MT400 or MT700 messages in any currency we code the outcome variable to nil. Column (4) and (5), as column (3) but averages treatment effects depending on whether a country is above or below the sample median in terms of its trade share with China. Columns (6) and (7) repeats the exercise for the intermediate import share, and columns (8) and (9) for the working capital of exports. See for exact definitions of these variables.

5.12 Table 7: Currency choice in payments with China

Read:

- Signing a swap line increases the RMB share of payments with China by 14.046 percentage points.
- The USD share falls by 7.898 percentage points.
- The EUR share falls by 2.627 percentage points.
- GBP/JPY/CHF and other international currencies also lose share.
- The home-currency share falls by 1.527 percentage points, but this estimate is not statistically significant.

Economic message. RMB adoption mainly replaces vehicle currencies, not local currencies. This matches the model's prediction that rising-currency pricing displaces dominant-currency pricing.

TABLE 7

Swap lines and currency choice in payments with China

	RMB	USD	EUR	GBP/JPY/CHF	Other	Home
	(1)	(2)	(3)	(4)	(5)	(6)
$\text{SwapLine}_{i,t}$	14.046^{***}	-7.898^{***}	-2.627^{**}	-0.493^{***}	-2.796^{**}	-1.527
	(2.29)	(2.49)	(1.04)	(0.09)	(1.25)	(1.19)
N treated	20	20	20	20	20	20
N control	93	93	93	93	93	93
T	97	97	97	97	97	97
Time f.e.	Yes	Yes	Yes	Yes	Yes	Yes
Country f.e.	Yes	Yes	Yes	Yes	Yes	Yes
Method	BJS(24)	BJS(24)	BJS(24)	BJS(24)	BJS(24)	BJS(24)

Notes: Standard errors clustered by country in parentheses * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$
Estimates of equation (3.1) with the outcome variable being the currency composition of payments made to China (inclusive of Hong Kong and Macau). A payment is defined by SWIFT message types MT 103 and MT 202. The treatment variable is a dummy variable indicating whether the country's central bank, as of month t, has ever signed a swap line agreement with the PBoC ($\text{SwapLine}_{i,t}$). Sample period is October 2010 to October 2018, Mongolia is excluded from the set of treated countries. BJS(24) refers to the DiD imputation method from Borusyak *et al.* (2024). Column (1): the outcome variable is the share of payments in RMB. Column (2)–(5): the outcome variable is the share of payments in USD; EUR; GBP, JPY, and CHF; all other currencies. The coefficients on columns (1)–(5) must, by definition, sum to nil. Column (6) is the share of payments in the currency of the counterparty.

6 Short summary

- The paper studies how a currency becomes international before it becomes dominant.
- The empirical setting is the PBoC's RMB swap-line network from 2009 to 2018.
- A swap line gives a foreign central bank access to RMB liquidity, allowing it to backstop local RMB lending.
- The main data are SWIFT cross-border payment messages by currency and country pair.
- The baseline sample contains 114 developing economies and 11,058 country-month observations.
- Signing a swap line raises the probability that a country uses RMB payments by about 12-14 percentage points.
- RMB payment values and shares also increase, with effects that grow over time.
- The mechanism is that swap lines reduce the volatility and right-tail risk of RMB borrowing costs.
- Synthetic RMB borrowing costs fall after a swap line is signed.
- During the 2015-16 offshore RMB funding shock, countries with swap lines maintain RMB use while countries without swap lines reduce it.
- RMB trade-finance messages increase after swap-line adoption.
- Effects are stronger in countries with more China trade, more intermediate imports, and higher export working-capital needs.
- The RMB mainly replaces USD and EUR use in payments with China, not the recipient country's home currency.

- The theoretical model explains these results through complementarity between the currency of working-capital liabilities and the currency of sticky-price invoicing.
- The central conclusion is that lender-of-last-resort policy can help move a currency across the threshold into international use.

7 Conclusion

7.1 Core conclusion

Bahaj and Reis show that central-bank liquidity policy can help jumpstart the international use of a currency. The PBoC's RMB swap lines are associated with a large increase in the probability and value of RMB payments abroad.

7.2 Economic intuition

A firm that finances imported inputs in RMB is more willing to invoice sales in RMB because matching cost and revenue currencies reduces profit volatility under sticky prices. But foreign firms and banks will not use RMB finance if RMB borrowing costs can spike. A swap line caps that tail risk. Once enough firms cross the threshold, RMB use can move from zero to positive and persist.

7.3 Takeaway

The paper's message is not that swap lines alone can dethrone the USD. Rather, they can move a currency from nonuse to international use when the issuing country is already close to the relevant thresholds: sufficiently large in trade, sufficiently stable in monetary policy, and able to provide credible liquidity support. The RMB's rise therefore resembles the early rise of the USD: trade finance and central-bank backstops are central to currency internationalization.